

PREMKUMAR M. PATEL

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EDUCATION

Sankalchand Patel University <i>B.Tech in Computer Engineering, CGPA: 7.80 / 10.0</i> - Coursework: Machine Learning, Artificial Intelligence, Deep Learning Applications, Database Management Systems, Operating Systems, Data Structures & Algorithms, Software Engineering, NLP, Compiler Design, GenAI - Organizer in the college technical club, managing hackathons and technical workshops	Visnagar, Gujarat, India 2023 – 2027
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TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript
Backend & APIs: FastAPI, Flask, RESTful APIs, Async I/O, Pydantic, Webhooks, HMAC Verification, Rate Limiting
AI & Agentic Systems: LangGraph, LangChain, Claude API, OpenAI API, RAG Pipelines, Tool/Function Calling, Prompt Engineering
Databases & Vector Search: PostgreSQL, Redis, SQLite, FAISS, Qdrant, Pinecone
System Design & Scalability: Docker, Microservices, Message Queues (Celery), Load Balancing, Caching Strategies, API Gateways, CI/CD, Nginx
DevOps & Tools: Git, GitHub, Linux/Bash, Vercel, Render, Postman

FEATURED PROJECTS

AgentPay <i>Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Razorpay API</i> - Built an autonomous buyer-agent system where the LLM proposes purchases and a separate deterministic policy engine gates every payment before it reaches the Razorpay API - Logged every state change in an append-only audit trail backed by database triggers, with merchant data isolated per tenant and hard spend limits per consumer - Set up a sandbox-to-production pipeline with webhook-driven agent integration, HMAC signature verification, and automatic retry with backoff	Aug 2026 Live Demo GitHub
CampusCircle <i>Next.js, FastAPI, PostgreSQL, LLaMA 3.3 70B, TypeScript</i> - Designed and deployed the full backend and AI infrastructure for a verified student platform, scaling it solo to over 200 active campus users with average API response times under 150ms - Shipped three production AI features on LLaMA 3.3 70B: an automated support agent ("Reva"), an adaptive quiz engine that adjusts question difficulty in real time, and a YouTube-to-study-notes summarizer - Placed in the top 18 of over 500 participants nationwide in The Maverick Effect AI Challenge, run by the Dewang Mehta Foundation Trust	Nov 2025 – Jan 2026 Live Demo GitHub
CliniqueAI <i>Python, Flask, PostgreSQL, Scikit-learn, XGBoost, LangChain, Pydantic</i> - Trained and benchmarked Logistic Regression, Random Forest, and XGBoost classifiers for clinical diabetes risk prediction, tuning the decision boundary for explainability and high recall - Built a RAG-based patient copilot grounded on localized EHR records, with transactional CRUD endpoints, Pydantic data schemas, and physician notification alerts - Removed cold-start latency on Render's free tier by setting up an automated heartbeat with UptimeRobot	Jan 2026 – Feb 2026 Live Demo GitHub

HONORS

National AI Hackathon (Unstop) — Top 20 Finalist, out of 1,600+ teams <i>Selected among the top 20 software teams nationwide for CliniqueAI, a diagnostic copilot</i>	National Level Feb 2026
The Maverick Effect AI Challenge — Top 18 Finalist, out of 500+ participants <i>Recognized by the Dewang Mehta Foundation Trust for CampusCircle, an AI community platform</i>	National Level 2026

CERTIFICATIONS

HackerRank Software Engineer Certification <i>Problem Solving, SQL, REST APIs</i>	2025
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